Powertrain Control HLD

Synthetic AUTOSAR-inspired High-Level Design Knowledge Document

Important: This document is fictional, independently created for academic evaluation, and contains no confidential OEM data or reproduced AUTOSAR standard text.

Field	Value
Document ID	SYN-HLD-PT-001
Version	1.1
Status	Synthetic review candidate
Classification	Synthetic academic test data
Intended use	RAG, OCR, extraction, comparison, and evaluation
Authority	Not valid for production design or vehicle release

1. Revision purpose

Version 1.1 updates the synthetic powertrain baseline to introduce cruise control integration, renamed thermal terminology, and revised timing. It is intended for revision-comparison testing.

2. Change record

Change ID	Change type	Affected item	Description
CHG-101	Added	CruiseControlManager	New component provides If_CruiseTorqueDemand
CHG-102	Modified	If_ThermalLimit	Data element DeratingReason renamed ThermalLimitReason
CHG-103	Modified	Vehicle speed timing	Publication changed from 20 ms to 10 ms
CHG-104	Removed	DiagnosticEventManager consumer	Removed from If_EngineState consumer list
CHG-105	Modified	TorqueArbitrator	Now requires If_CruiseTorqueDemand

3. Updated component catalogue

Component ID	Software Component	Responsibility	Provided Interface	Required Interface
SWC-PT-001	EngineControlCoordinator	Coordinates torque and engine operating states	If_EngineState	If_VehicleSpeed, If_PedalPosition
SWC-PT-002	VehicleStateManager	Publishes normalized vehicle speed and motion state	If_VehicleSpeed	If_WheelSpeedRaw
SWC-PT-003	PedalPositionAdapter	Validates accelerator pedal channels	If_PedalPosition	If_AdcPedal
SWC-PT-004	TorqueArbitrator	Combines driver and supervisory torque requests	If_TorqueRequest	If_PedalPosition, If_EngineState
SWC-PT-005	ThermalProtectionManager	Applies torque derating from temperature	If_ThermalLimit	If_CoolantTemperature
SWC-PT-006	CruiseControlManager	Generates supervised cruise torque demand	If_CruiseTorqueDemand	If_VehicleSpeed, If_DriverOverride

4. Updated interfaces

Interface ID	Interface Name	Type	Provider	Consumers	Key Data Elements
IF-001	If_VehicleSpeed	Sender-Receiver	VehicleStateManager	EngineControlCoordinator, CruiseControlManager	VehicleSpeed_kph, MotionState
IF-002	If_PedalPosition	Sender-Receiver	PedalPositionAdapter	EngineControlCoordinator, TorqueArbitrator	PedalPosition_pct, PedalValidity
IF-003	If_EngineState	Sender-Receiver	EngineControlCoordinator	TorqueArbitrator	EngineRunning, EngineSpeed_rpm
IF-004	If_TorqueRequest	Client-Server	TorqueArbitrator	EngineControlCoordinator	RequestedTorque_Nm, RequestSource
IF-005	If_ThermalLimit	Sender-Receiver	ThermalProtectionManager	EngineControlCoordinator	MaximumTorque_Nm, ThermalLimitReason
IF-006	If_CruiseTorqueDemand	Sender-Receiver	CruiseControlManager	TorqueArbitrator	CruiseTorque_Nm, CruiseState

5. Updated behaviour

VehicleStateManager publishes VehicleSpeed_kph every 10 ms. CruiseControlManager consumes vehicle speed and produces CruiseTorque_Nm. TorqueArbitrator selects between driver demand and cruise demand according to CruiseState and driver override.

6. Known review observation

The input If_DriverOverride is referenced by CruiseControlManager but is intentionally not defined in this revision. This controlled defect is included for completeness-check evaluation.

7. Approval note

This is a synthetic revision comparison artifact. It shall not be treated as an authoritative vehicle architecture.